

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**/Supplementary Winter Examination – 2024****Branch : Food Engineering & technology****/Food Technology and management****Semester :VI****Subject Code & Name: BTFT601 Mass Transfer II****Course: T.Y.B.Tech.****Max Marks: 60****Date 18/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	What is mass transfer? a) Transfer of momentum b) Movement of energy across phases c) Movement of mass due to a concentration difference d) None of the mentioned		1
2	Which among the following is always true for mass transfer to occur? a) Difference in concentration b) Difference in chemical potential c) Difference in pressure d) Difference in temperature		1
3	The mass transfer by convection can take place under how many conditions? a) 3 b) 4 c) 2 d) 1		1
4	What is the unit of diffusivity? a) m/s b) m ³ /s c) m ² /s d) m/s ²		1
5	What causes the concentration at the interphase to be unequal, according to Lewis and Whitman theory?		1

	a) Moderate mass transfer rate	b) Low mass transfer rates	c) High mass transfer rates	d) None of the mentioned		
6	What will be poor if the holding time is longer in a mixer?					1
	a) Momentum transfer	b) Mass transfer	c) Heat transfer	d) Both mass and momentum transfer		
7	In humidification, where must the gas be for mass transfer to occur?					1
	a) Soluble	b) Insoluble	c) Partially soluble	d) Inert		
8	Which term represents the enhancement of mass transfer in turbulent flow?					1
	a) Eddy diffusivity	b) Laminar flow	c) Molecular diffusion	d) Surface tension		
9	. Which process utilizes gas absorption as a mass transfer operation					1
	a) Ammonia production	b) Filtration	c) Sedimentation	d) Water desalination		
10	What is the significance of the concentration gradient in mass transfer?					1
	a) It acts as the driving force	b) None of the mentioned	c) It does not affect the process	d) It slows down the process		
11	Which of the following is NOT a mass transfer operation?					1
	a) Gas absorption	b) Heat conduction	c) Drying	d) Distillation		
12	Which of these is an example of mass transfer?					1
	a) Expansion of gas	b) Heat conduction in a rod	c) Freezing of water	d) Mixing of two liquids		
Q. 2	Solve the following.					12

A)	Selection of absorbent & limiting liquid gas ratio		6
B)	Explain preliminaries and molecular distillation.		6
Q.3	Solve the following.		12
A)	Explain azeotropic and extractive distillation.		6
B)	Explain principle and applications of steam distillation.		6
	Discuss loading and flooding in packed towers.		
Q. 4	Solve Any Two of the following.		12
A)	Derive an expression for flux N_A in equimolar counter diffusion and show the graphical representation		6
B)	Difference between flash & differential distillation for binary system.		6
C)	Explain Raoult's law and Henry law & its uses.		6
Q.5	Solve Any Two of the following.		12
A)	Selection criteria for solvent.		6
B)	Explain concept of transfer unit for the design of packed absorbers		6
C)	Limitations & application for distribution		6
Q. 6	Solve Any Two of the following.		12
A)	Explain dew point & bubble point estimation for binary distillation.		6
B)	Explain Equilibria for absorption system & its uses.		6
C)	Explain Swenson – Walker & vacuum crystallizers & crystallization and its role.		6
	*** End ***		